

ZERO MODES ON THE LATTICE. THE VORTEX FERMION SYSTEM

Leticia F. CUGLIANDOLO ^{a,b}, Eduardo FRADKIN ^{c,a} and Fidel A. SCHAPOSNIK ^a

^a *Departamento de Física, Universidad Nacional de La Plata, 1900 La Plata, Argentina*

^b *Departamento de Física, Universidad Nacional de Mar del Plata, 7600 Mar del Plata, Argentina*

^c *Department of Physics, University of Illinois at Urbana-Champaign, 1110 West Green Street, Urbana, IL 61801, USA*

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We investigate the zero mode problem for the lattice Dirac equation in the background of an abelian vortex. Following Jackiw–Rossi analysis in the continuum, we include a fermion–scalar field interaction and we show that the Dirac equation for (staggered) lattice fermions possesses $|N|$ zero eigen-value modes in the N -vortex background. Our result suggests the validity of an index theorem in a two-dimensional lattice.

Topologically non-trivial classical configurations with finite energy have been playing a role of increasing importance in quantum field theory. Scalar kinks, vortices, monopoles, find relevant applications in particle physics [1], cosmology [2] and also in condensed matter physics [3].

In the presence of topologically non-trivial configurations, quantum systems may exhibit unexpected and interesting behaviors like charge fractionization of fermions moving in the background of a scalar kink or a magnetic monopole [4], parity violation [2], etc. These phenomena are closely related to the appearance of anomalies and the existence of zero energy solutions to the Dirac equation, features which are also crucial in understanding the so-called $U_A(1)$ problem in QCD [5–7], properties of superconducting cosmic strings [8], etc.

Since the role played by non-trivial configurations is intrinsically non-perturbative, it is natural to investigate their relevance when field theories are defined on a lattice as a way to go beyond perturbation theory ^{#1}. In particular, in view of the success of the “topological resolution” for the $U_A(1)$ problem – made quantitative by the Witten–Veneziano [6–8] formula based on the index theorem [10] – it is interesting to determine whether zero-modes properties remain valid for lattice gauge theories.

However, serious difficulties arise in trying to establish a connection between topological charge and the number of zero modes of the Dirac equation (Atiyah–Singer index theorem) in lattice gauge theories [9]. These difficulties are related to the definition of an appropriate topological charge, the problems posed by lattice fermions, etc. [9].

Using numerical methods, exact and approximate zero modes have been found for QED_2 and the remnants of the index theorem on the lattice have been discussed [11] (see also refs. [12,13]).

In the present paper we study an analogous problem without resorting to numerical methods. Indeed, we present an exact solution for the zero modes of a system of (staggered) lattice fermions in the background of an abelian vortex. In the continuum this problem was solved by Jackiw and Rossi [14] who found $|N|$ zero eigen-value modes in an N -vortex background. These authors included a scalar–fermion coupling due to the prominent role the Higgs field plays in the determination of the properties of the Dirac equation solutions [13]. We have been able to extend the Jackiw–Rossi Ansatz to the lattice, thus constructing exact zero mode solutions.

^{#1} See ref. [9] for a review.

As in the continuum we show that in the N -vortex sector there are $|N|$ isolated, nearly independent lattice zero modes. In accordance with the Atiyah–Singer theorem, for positive (negative) N , zero modes have only upper (lower) components. As in the Jackiw–Rossi example [14], the presence of the Higgs field is also crucial in the lattice theory for the existence and properties of zero modes.

We use a lattice version in which time is left continuous [15], replacing two-space by a simple square lattice with spacing a . As we stated above, we shall work with staggered fermions [16] in the presence of a vortex. The Dirac spinors and matrices can be taken with only two components [14]. For the lattice Dirac hamiltonian we take

$$\begin{aligned}
 H = \sum_r \{ & (i/2a) \{ \phi^\dagger(\mathbf{r}) \exp[-i\theta_1(\mathbf{r})] \phi(\mathbf{r} + a\hat{e}_1) - \phi^\dagger(\mathbf{r} + a\hat{e}_1) \exp[+i\theta_1(\mathbf{r})] \phi(\mathbf{r}) \} \\
 & + (1/2a) \{ \phi^\dagger(\mathbf{r}) \exp[-i\theta_2(\mathbf{r})] \phi(\mathbf{r} + a\hat{e}_2) + \phi^\dagger(\mathbf{r} + a\hat{e}_2) \exp[+i\theta_2(\mathbf{r})] \phi(\mathbf{r}) \} (-1)^{x+y} \\
 & + \frac{1}{2} ig [\Phi(\mathbf{r}) \phi^\dagger(\mathbf{r}) \{ \phi^\dagger(\mathbf{r} + a\hat{e}_1) \exp[+i\theta_1(\mathbf{r})] + \phi^\dagger(\mathbf{r} - a\hat{e}_1) \exp[+i\theta_{-1}(\mathbf{r})] \\
 & + \phi^\dagger(\mathbf{r} + a\hat{e}_2) \exp[+i\theta_2(\mathbf{r})] + \phi^\dagger(\mathbf{r} - a\hat{e}_2) \exp[+i\theta_{-2}(\mathbf{r})] \}] (-1)^{x+y} + \text{h.c.} \} , \quad \mathbf{r} = (ma, na) . \quad (1)
 \end{aligned}$$

The field, $\phi(t, \mathbf{r})$ is a single component (staggered) fermion field. In the continuum limit it becomes equivalent to two species of two-component Dirac fermions ψ . In this limit only one species with upper component ψ_1 and lower component ψ_2 (equal to the average of the ϕ field on even and odd sublattice sites respectively) couples to the scalar field Φ . The other species remains massless and decoupled.

The gauge field is represented by the link variable $\theta_\mu(\mathbf{r})$, connected with the continuum gauge field $A_\mu(\mathbf{r})$ by $\theta_\mu(\mathbf{r}) = aeA_\mu(\mathbf{r})$, where e is its charge. The scalar field Φ has charge $2e$.

We choose for $\theta_\mu(\mathbf{r})$ a static $U(1)$ -vortex configuration with N units of magnetic flux [15] at a site $\mathbf{R}$ of the dual lattice.

$$\oint_r \theta_\mu(\mathbf{r}) = 2\pi N, \quad N \in \mathbb{Z}, \quad (2)$$

where $\oint$ indicates a sum over *any* closed contour on the lattice, including the vortex at $\mathbf{R}$ in its interior.

We have also included in eq. (1) a scalar-fermion coupling (with coupling constant g) since experience in the continuum shows that binding for topologically interesting zero-eigenvalue modes is provided by the scalar Higgs field, which, in fact, is necessary in obtaining a regular vortex solution [17]. The Higgs field has been taken as

$$\Phi(\mathbf{r}) = \exp(iN\phi)f(\mathbf{r}). \quad (3)$$

The choice we have made for the coupling between the fermions and the scalar field guarantees the consistency of the staggered-fermion equation of motion, which takes the form

$$\begin{aligned}
 -\partial(\mathbf{r})/\partial t = & (\sigma_1/2a) \{ \psi(\mathbf{r} + a\hat{e}_1) \exp[-i\theta_1(\mathbf{r})] - \psi(\mathbf{r} - a\hat{e}_1) \exp[-i\theta_{-1}(\mathbf{r})] \} \\
 & + (\sigma_2/2a) \{ \psi(\mathbf{r} + a\hat{e}_2) \exp[-i\theta_2(\mathbf{r})] - \psi(\mathbf{r} - a\hat{e}_2) \exp[-i\theta_{-2}(\mathbf{r})] \} \\
 & - \frac{1}{2} ig \sigma_2 \{ [\Phi(\mathbf{r} + a\hat{e}_1) \psi^\dagger(\mathbf{r} + a\hat{e}_1) \exp[-i\theta_1(\mathbf{r})] + \Phi(\mathbf{r} - a\hat{e}_1) \psi^\dagger(\mathbf{r} - a\hat{e}_1) \exp[-i\theta_1(\mathbf{r})] \\
 & + \Phi(\mathbf{r} + a\hat{e}_2) \psi^\dagger(\mathbf{r} + a\hat{e}_2) \exp[-i\theta_2(\mathbf{r})] + \Phi(\mathbf{r} - a\hat{e}_2) \psi^\dagger(\mathbf{r} - a\hat{e}_2) \exp[-i\theta_{-2}(\mathbf{r})]] \\
 & + \Phi(\mathbf{r}) [\psi^\dagger(\mathbf{r} + a\hat{e}_1) \exp[-i\theta_1(\mathbf{r})] - \psi^\dagger(\mathbf{r} - a\hat{e}_1) \exp[-i\theta_{-1}(\mathbf{r})] + \psi^\dagger(\mathbf{r} - a\hat{e}_2) \exp[-i\theta_{-2}(\mathbf{r})]] \} , \quad (4)
 \end{aligned}$$

where $\psi = (\psi_1, \psi_2)^T$ and σ_1, σ_2 are the Pauli matrices. In this formalism the other remaining massless, decoupled species of Dirac fermions are ignored.

Separation of the time variable requires the two-phase Ansatz

$$\psi(t, \mathbf{r}) = \exp(-iEt) \psi^{(+)}(\mathbf{r}) + \exp(iEt) \psi^{(-)}(\mathbf{r}). \quad (5)$$

As in the continuum [14], the transformation $\psi^{(+)} \rightarrow \sigma_3 \psi^{(+)}$, $\psi^{(-)} \rightarrow \sigma_3 \psi^{(-)}$, takes a solution with eigenvalue E into one with eigenvalue $(-E)$.

For zero energy, the two equations satisfied by $\psi^{(+)}$, $\psi^{(-)}$ collapse into one:

$$\begin{aligned} 0 = & (\sigma_1/2a) \{ \psi(\mathbf{r} + a\hat{e}_1) \exp[-i\theta_1(\mathbf{r})] - \psi(\mathbf{r} - a\hat{e}_1) \exp[-i\theta_{-1}(\mathbf{r})] \} \\ & + (\sigma_2/2a) \{ \psi(\mathbf{r} + a\hat{e}_2) \exp[-i\theta_2(\mathbf{r})] - \psi(\mathbf{r} - a\hat{e}_2) \exp[-i\theta_{-2}(\mathbf{r})] \} \\ & - (i\sigma_1/2a) \{ [\Phi(\mathbf{r} + a\hat{e}_1) \psi^\dagger(\mathbf{r} + a\hat{e}_1) \exp[-i\theta_1(\mathbf{r})] + \Phi(\mathbf{r} - a\hat{e}_1) \psi^\dagger(\mathbf{r} - a\hat{e}_1) \exp[-i\theta_1(\mathbf{r})] \\ & + \Phi(\mathbf{r} + a\hat{e}_2) \psi^\dagger(\mathbf{r} + a\hat{e}_2) \exp[-i\theta_2(\mathbf{r})] + \Phi(\mathbf{r} - a\hat{e}_2) \psi^\dagger(\mathbf{r} - a\hat{e}_2) \exp[-i\theta_{-2}(\mathbf{r})]] \\ & + \Phi(\mathbf{r}) [\psi^\dagger(\mathbf{r} + a\hat{e}_1) \exp[-i\theta_1(\mathbf{r})] - \psi(\mathbf{r} - a\hat{e}_1) \exp[-i\theta_{-1}(\mathbf{r})] + \psi^\dagger(\mathbf{r} - a\hat{e}_2) \exp[-i\theta_{-2}(\mathbf{r})]] \} . \end{aligned} \quad (6)$$

Now, this zero energy equation residing on two-dimensional space can be solved by the Ansatz

$$\begin{aligned} & (\pm \sigma_j/2a) \exp[-i\theta_{\pm j}(\mathbf{r})] \psi(\mathbf{r} \pm a\hat{e}_j) - \frac{1}{8} i g \sigma_2 \{ \Phi(\mathbf{r} \pm a\hat{e}_j) \exp[-i\theta_{\pm j}(\mathbf{r})] + \Phi(\mathbf{r}) \exp[-i\theta_{\pm j}(\mathbf{r})] \} \psi^\dagger(\mathbf{r} \pm a\hat{e}_j) \\ & = \exp[\sigma_3 \Omega(\mathbf{r})] [\pm (\sigma_j/2a) \chi(\mathbf{r} \pm a\hat{e}_j) - i g \sigma_2 \Phi(\mathbf{r}) \chi^\dagger(\mathbf{r} \pm a\hat{e}_j)] , \end{aligned} \quad (7)$$

which eliminates $\theta_\mu(\mathbf{r})$ from eq. (6).

The real function $\Omega(\mathbf{r})$ is related to the vortex through the equation

$$\lim_{a \rightarrow 0} \frac{1}{e} \varepsilon_{ij} A'_j \Omega(\mathbf{r}) = A_i(\mathbf{r}) , \quad (8)$$

with

$$A_i \psi = (1/2a) [\psi(\mathbf{r} + a\hat{e}_i) - \psi(\mathbf{r} - a\hat{e}_i)] . \quad (9)$$

We propose the following form for sums and differences of nearest neighbors on each sublattice:

$$\begin{aligned} & \frac{1}{4} [\chi_1(\mathbf{r} + a\hat{e}_1) + \chi_1(\mathbf{r} + a\hat{e}_2) + \chi_1(\mathbf{r} - a\hat{e}_1) + \chi_1(\mathbf{r} - a\hat{e}_2)] = U_1(r) \exp(im\theta) + V_1(r) \exp[i(n-1-m)\theta] , \\ & (1/2a) [\chi_1(\mathbf{r} + a\hat{e}_1) - \chi_1(\mathbf{r} - a\hat{e}_1)] + (i/2a) [\chi_1(\mathbf{r} + a\hat{e}_2) - \chi_1(\mathbf{r} - a\hat{e}_2)] \\ & = \exp[-i(m+1)\theta] [\Delta_r U_1(r) - (m/r) U_1(r)] + \exp[i(n+m)\theta] \{ \Delta_r V_1(r) - [(n-1-m)/r] V_1(r) \} , \end{aligned} \quad (10)$$

$$\begin{aligned} & \frac{1}{4} [\chi_2(\mathbf{r} + a\hat{e}_1) + \chi_2(\mathbf{r} + a\hat{e}_2) + \chi_2(\mathbf{r} - a\hat{e}_1) + \chi_2(\mathbf{r} - a\hat{e}_2)] = U_2(r) \exp(-im\theta) + V_2(r) \exp[i(n-1-m)\theta] , \\ & (1/2a) [\chi_2(\mathbf{r} + a\hat{e}_1) - \chi_2(\mathbf{r} - a\hat{e}_1)] - (i/2a) [\chi_2(\mathbf{r} + a\hat{e}_2) - \chi_2(\mathbf{r} - a\hat{e}_2)] \\ & = \exp[i(m+1)\theta] [\Delta_r U_2(r) - (m/r) U_2(r)] + \exp[i(n-m)\theta] \{ -\Delta_r V_2(r) - [(n+1+m)/r] V_2(r) \} , \end{aligned} \quad (11)$$

with $m \in \mathbb{Z}$, in order to ensure single-valuedness.

In eqs. (10), (11), we have introduced an operator Δ_r playing the role of a lattice gradient, from which we must recover ∂_r , in the $a \rightarrow 0$ limit. We shall construct it as a power series in a , in terms of cartesian finite differences, in such a way that $\Delta_r f(r)$ should depend only on the radial coordinate r .

We then write

$$\Delta_r = \sin \theta \left(\Delta_2 + \sum_{n \geq 3} C_n \Delta_2^{(n)} a^{n-1} \right) + \cos \theta \left(\Delta_1 + \sum_{n \geq 3} C'_n \Delta_1^{(n)} a^{n-1} \right) , \quad (12)$$

where Δ_i ($i = 1, 2$) is given by eq. (9), and the other terms in the sum are calculated iteratively in a straightforward way.

In particular,

$$\Delta_2^{(3)} g(y) = [\frac{3}{4} g(y+2a) - \frac{1}{4} g(y-2a) - 2g(y+a) + \frac{3}{2} g(y)] / a^3 , \quad (13)$$

and $C_3 = \frac{1}{6}$.

From the Ansatz (10), (11) one gets for the upper component equation

$$\Delta_r U_1(r) - (m/r)U_1(r) + gf(r)V_1^*(r) = 0, \quad \Delta_r V_1(r) - [(n-1-m)/r]V_1(r) + gf(r)U_1^*(r) = 0, \quad (14)$$

with Δ_r defined by eq. (12).

Two similar equations arise for the lower component (they can be obtained from eq. (14) changing N into $-N$).

It is easy to prove that the Ansatz (10), (11) is consistent and unique.

Decomposing $U_1(r)$ and $V_1(r)$ into real and imaginary parts, one can see that these functions can be written in the form

$$U_1(r) = \exp(i\alpha_m)\mu_m(r), \quad V_1(r) = \exp(-i\alpha_m)V_m(r), \quad (15)$$

where $\mu_m(r)$ and $V_m(r)$ are real functions also satisfying eq. (14).

Demanding $\mu_m(r)$ and $V_m(r)$ to be well-behaved near the origin one gets, as in the continuum limit, the following condition obtained for m :

$$N-1 \geq m \geq 0. \quad (16)$$

Thus, for positive vortex number N , solutions with only the upper component exist; for negative vortex number $|N|$, solutions with only the lower component exist.

Our explicit construction shows that if we call n_+ (n_-) the number of zero modes with only upper (lower) component, the following relation holds:

$$N = n_+ - n_-. \quad (17)$$

This relation between the topological charge N and the zero modes can be interpreted as a check of the lattice version of the Atiyah–Singer theorem. (Notice that, as we explained above, flavor degeneracy for staggered fermions is broken due to the presence of the scalar–fermion coupling.)

Numerical analysis [11], either with staggered or Wilson fermions, in the two-dimensional lattice $U(1)$ model, has also proved the existence of approximate and exact zero modes related to the gauge field topological charge through eq. (17).

It has also been shown that there exists a definition of a certain quantity $\tilde{Q}$ [12,13] in terms of fermion fields, which in the limit a going to zero, gives the topological charge Q ($Q=N$ in our case). It is remarkable that in the model we have analyzed, zero modes satisfying relation (17) can be constructed explicitly for arbitrary e and g (all coupling constants appearing in the model) and for all values of the lattice spacing.

Note that even in the continuum, the model we have considered is defined in a non-compact ($\mathbb{R}^2$) and hence the Atiyah–Singer theorem cannot be applied naively.

We conclude by noting that if the condition on m is weakened, demanding fermion bilinears to be single-valued, $2m \in \mathbb{Z}$, more zero energy modes exist, and relation (17) is no longer valid.

In summary, we have explicitly constructed the exact zero modes for staggered fermions in a vortex background. We have checked that an analog of the index theorem applies on the lattice.

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References

- [1] See, e.g., C. Rebbi and G. Soliani, eds., *Solitons and particles* (World Scientific, Singapore, 1985).

- [2] See, e.g., K.A. Olive, in: Grand unification with and without supersymmetry, eds. C. Kounas et al., International School of Advanced Studies (Trieste) Lecture Series #2 (World Scientific, Singapore, 1986).
- [3] See, e.g., Topology in field theory, Proc. Santa Fe American Physical Society Meeting (1984); E. Fradkin, in: ELAF '87, eds. J. Giambiagi et al. (World Scientific, Singapore, 1988).
- [4] R. Jackiw and C. Rebbi, Phys. Rev. D 13 (1976) 3398.
- [5] G. 't Hooft, Phys. Rev. Lett. 37 (1976) 8; Phys. Rev. D 14 (1976) 3432.
- [6] E. Witten, Nucl. Phys. B 156 (1979) 269.
- [7] G. Veneziano, Nucl. Phys. B 159 (1979) 213.
- [8] E. Witten, Nucl. Phys. B 249 (1985) 557.
- [9] See, e.g., A.S. Kronfield, in: Topological aspects of lattice gauge theories, Proc. LAT 87 (Seillac, France), DESY Report No. 87-152, unpublished, and references therein.
- [10] M. Atiyah and I. Singer, Ann. Math. 87 (1968) 484.
- [11] J. Smit and J.C. Vink, Nucl. Phys. B 286 (1987) 485; B 298 (1988) 557; ITFA Reports #87-15, #88-2, unpublished.
- [12] I.A. Fox, M.L. Laursen, G. Schierhdz, J.P. Gilchrist and M. Gockeler, Phys. Lett. B 158 (1985) 332.
- [13] F. Karsch, E. Seiler and I.O. Stamatescu, Nucl. Phys. B 271 (1986) 346.
- [14] R. Jackiw and P. Rossi, Nucl. Phys. B 190 [FS3] (1981) 681.
- [15] H. Kosterlitz and D.J. Thouless, J. Phys. C 6 (1973) 1181.
- [16] See, e.g., L. Susskind, in: Weak and electromagnetic interactions in high energy (Les Houches, 1976), eds. R. Balian et al. (North-Holland, Amsterdam, 1977).
- [17] H.B. Nielsen and P. Olesen, Nucl. Phys. B 61 (1973) 45.